

Activity Diagram

Project: LearnLens AI

1. End-to-End Multimodal Observation & Tutoring Lifecycle

```
stateDiagram-v2
[*] --> InitializeSession: User clicks 'Start Learning'

state "Capture & Stream Layer" as Capture {
InitializeSession --> SelectSource: Chrome Native DisplayMedia Picker
SelectSource --> StreamActive: User authorizes Tab/Window
StreamActive --> AudioFork: Split Stream
StreamActive --> VideoFork: Split Stream
}

state "Perception & Processing" as Processing {
AudioFork --> TranscribeAudio: faster-whisper (CPU / GPU)
TranscribeAudio --> SynchronizeTimestamps: Attach 00:00 start/end

VideoFork --> ComputeDHash: Sample Frame (3000ms)
ComputeDHash --> FilterDuplicate: Compare with previous hash

state duplicate_decision <<choice>>
FilterDuplicate --> duplicate_decision
duplicate_decision --> DropFrame: Hamming dist < 8
duplicate_decision --> AnalyzeFrame: Hamming dist >= 8

AnalyzeFrame --> RunOCR: Extract visible labels & text
AnalyzeFrame --> ClassifyVisual: Tag as DIAGRAM / TABLE / SLIDE

SynchronizeTimestamps --> FuseKnowledge: Aligned Multi-modal Fusion
RunOCR --> FuseKnowledge
ClassifyVisual --> FuseKnowledge
}

state "Knowledge & RAG Storage" as Knowledge {
FuseKnowledge --> ExtractConcepts: Discover Definitions & Mechanisms
ExtractConcepts --> GenerateSubwordVectors: Unit Vector Projection
GenerateSubwordVectors --> CommitToSessionScope: Save to SQLite & Vectors
}

state "Human Interaction & Tutoring" as Tutoring {
CommitToSessionScope --> ReadyForUser: Dashboard Updated
ReadyForUser --> ReceiveQuestion: User prompts AI Teacher
ReceiveQuestion --> RetrieveGrounding: Cosine Sim + BM25 in Session RAG
RetrieveGrounding --> SynthesizeAnswer: Prompt with 12 Pedagogical Modes
SynthesizeAnswer --> DisplayEvidence: Cite Timestamps & Slide Thumbnails
DisplayEvidence --> ReadyForUser: Await Next Doubt
}

state "Study Artifact Generation" as Exports {
ReadyForUser --> TriggerPPT: User clicks 'Generate PPT'
TriggerPPT --> LayoutSlides: python-pptx 16:9 Deck + Diagrams
LayoutSlides --> DownloadPPTX: Deliver .pptx file

ReadyForUser --> TriggerPDF: User clicks 'Generate PDF'
TriggerPDF --> FormatDocument: ReportLab Visual Pack / Teaching Report
FormatDocument --> DownloadPDF: Deliver .pdf file
```

